

Foundation model

💡 Idea

- A foundation model is any model trained on broad data, generally by self-supervision at scale, that can be adapted (for example by [fine-tuning](#) or prompting) to a wide range of downstream tasks — one model reused as the base for many tasks instead of a separate model trained per task ([Bommasani2021OnTO](#); [sec. 1, p. 3](#)).

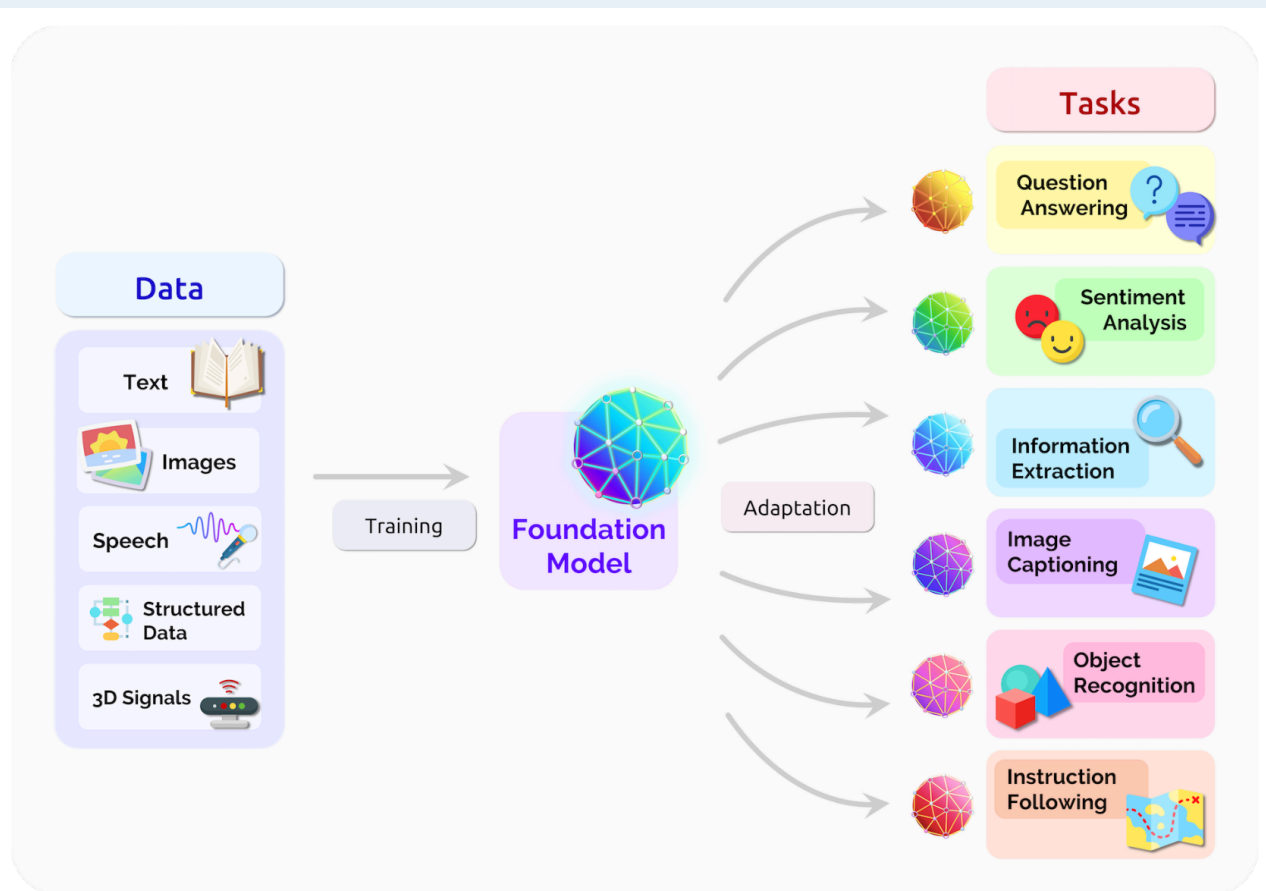


Fig. 2. A foundation model can centralize the information from all the data from various modalities. This one model can then be adapted to a wide range of downstream tasks.

- The technique is not new — deep neural networks and self-supervised learning are decades old — but the scale and scope are, which is what the term marks ([Bommasani2021OnTO](#); [sec. 1, p. 3](#)).
- It comes in variants by input type: a text-only large language model such as [GPT-3](#), and a multimodal [vision-language model](#) that also takes images as input ([Zhai2024FineTuningLV](#); [sec. 1, p. 2](#)).

🔍 Why It Matters

- Two properties are what make the paradigm significant ([Bommasani2021OnTO](#); [sec. 1.1, p. 3](#)). The first is [emergence](#) — behaviour implicitly induced rather than explicitly constructed, such as [in-context learning](#) arising unanticipated from scale.
- The second is [homogenization](#) — many applications consolidating onto a few shared models, which gives strong leverage but creates a single point of failure where the base model's defects are inherited downstream ([Bommasani2021OnTO](#); [sec. 1.1, p. 3](#)).
- It separates one expensive pretraining run from task-solving: the same model is adapted to many tasks cheaply rather than rebuilt per task, the shift away from the older task-specific-model paradigm ([Brown2020LanguageMA](#); [sec. 1, p. 1](#)).

⊗ Not This

- Not a task-specific model trained from scratch on one task's labelled dataset; the foundation-model approach replaces that per-task design with one reusable pretrained base ([Brown2020LanguageMA](#); [sec. 1, p. 1](#)).
- Not an [LLM agent](#): a foundation model is the underlying model, while an [LLM agent](#) wraps one in profiling, memory, planning, and action to sense and act in an environment over time.

☰ Examples

- [GPT-3](#), a 175-billion-parameter autoregressive language model pretrained on filtered CommonCrawl plus higher-quality WebText, book, and Wikipedia corpora, then applied to many tasks with no weight updates ([Brown2020LanguageMA](#); [abstract, p. 1](#); [sec. 2.2, p. 3](#)).
- A [vision-language model](#) such as the LLaVA backbone, or the commercial GPT-4V and Gemini, used as a general-purpose agent that reasons over an image and a text prompt ([Zhai2024FineTuningLV](#); [sec. 1, p. 2](#)).

⚠ Contradictions

- None yet

Disconfirming Evidence

- "Foundation" should not be read as broad competence in every domain: [GPT-3's](#) authors note that large pretrained language models are not grounded in non-text experience such as video or physical interaction, and so "lack a large amount of context about the world" ([Brown2020LanguageMA](#); [sec. 5, p. 8](#)).

Open Questions

- The boundary is fuzzy: the defining source sets no scale threshold or sharp line against an ordinary large pretrained model, so what counts as a foundation model is a matter of degree and significance, not a crisp test ([Bommasani2021OnTO](#); [sec. 1.1.1, p. 6](#)).
- Whether broad pretraining genuinely acquires reusable general skills or largely memorizes patterns later recognized at inference; the scale-driven gains are documented without a mechanism.

Connections

- The base an [LLM agent](#) uses as its core controller.
- Adapted to downstream tasks along the [in-context-learning to fine-tuning spectrum](#) — from [in-context learning](#) (no weight change), through [parameter-efficient fine-tuning](#) and [fine-tuning](#), to [end-to-end RL fine-tuning](#).
- Its two defining properties have their own pages: [emergence](#) and [homogenization](#); the text-only versus multimodal split is a plain language model versus a [vision-language model](#), and its scale-driven gains tie to [scaling](#).
- The base unit (rung 0) of the [foundation model to multi-agent system progression](#), which traces how the design problem relocates from model capability to team coordination as the unit grows.

Sources

- [Bommasani2021OnTO](#)
- [Brown2020LanguageMA](#)

- [Zhai2024FineTuningLV](#)